JAMES MORRISON

Research Engineer

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PROFESSIONAL SUMMARY

I'm a Research Engineer with three years of experience building machine learning systems and developing novel algorithms for real-world applications. I've worked across computer vision and data analysis projects, with a particular interest in optimization techniques and statistical modelling. I prefer hands-on problem solving and have a track record of taking research prototypes through to implementation stages.

WORK EXPERIENCE

Research Engineer

DataFlow Systems

www.dataflowsystems.io

Edinburgh, United Kingdom

June 2022 - Present

Designed and implemented a real-time anomaly detection system for industrial sensor data, reducing false positives by 34% through custom feature engineering and ensemble methods.

Developed Python-based data pipeline infrastructure processing 2TB daily, integrating with existing PostgreSQL and Apache Spark environments.

Collaborated with product team to translate research findings into production-grade code, including comprehensive unit testing and documentation.

Conducted literature reviews and experimented with three different neural network architectures before selecting the optimal solution for our use case.

Machine Learning Developer

Insight Analytics Ltd

www.insightanalytics.co.uk

Edinburgh, United Kingdom

March 2021 - May 2022

Built classification models for customer behaviour prediction, achieving 89% accuracy and generating actionable business insights for the sales department.

Automated data preprocessing workflows using pandas and scikit-learn, reducing manual preparation time from 6 hours to 30 minutes per dataset.

Presented quarterly findings to stakeholders, explaining model limitations and confidence intervals in accessible terms.

Maintained Jupyter notebooks and created visualizations using matplotlib and Plotly to communicate results to non-technical audiences.

Junior Research Analyst

NeuralWorks Technologies

www.neuralworks.net

Glasgow, United Kingdom

July 2020 - February 2021

Assisted senior researchers in image classification experiments using TensorFlow and OpenCV on medical imaging datasets.

Performed statistical analysis on experimental results and contributed to two peer-reviewed papers on convolutional neural networks.

Documented experimental protocols and maintained organized version control using Git for collaborative research projects.

Troubleshot hardware issues with GPU clusters and helped optimize training times through batch processing improvements.

EDUCATION

Bachelor of Science in Computer Science

University of Edinburgh

Edinburgh, United Kingdom

Graduated June 2020

Grade: Upper Second Class Honours

TECHNICAL SKILLS

Languages: Python, SQL, Bash, Java

Machine Learning: TensorFlow, PyTorch, scikit-learn, XGBoost, ensemble methods, feature engineering

Data Processing: pandas, NumPy, Apache Spark, PostgreSQL

Visualization: Matplotlib, Plotly, Seaborn

Tools and Platforms: Git, Jupyter Notebook, Linux, Docker, AWS

Techniques: Statistical analysis, algorithm optimization, time series analysis, supervised and unsupervised learning